Analysis of regular and premium gasoline and their environmental impacts in Sulaymaniyah region

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ABSTRACT

Gasoline is one of the major sources of energy since the beginning of invention in automobile in 19th century, it was made from refining and distillation process of crude oil, since then it has been progressed in to many forms to increase its versatility and have less environmental impact when combusting by vehicles, by adding different additives to increase its octane value which is result to a better gasoline that can be used by citizens. In this study we observed gasoline types (premium and regular) in Sulaymaniyah city by taking samples from (7) stations in different location to examine them in a laboratory test station which was performed in Tanjero area near Sulaymaniyah city. Three different Device has been used for examining the gasoline components (premium and regular), (Eravap) for measuring vapor pressure, (Eralytics) multi-fuel analyzer which measure (Ron, Mon, AKI, Benzene, Aromatics, Toluene, MTBE, and Oxygen, Ethanol, Oxygenate, Olefins, (IBP, FBP, T10, T50, T90), and (Zetlex) octane tester. Then Portable gas analyzer-series 500 model with three sensor heads (CO2, CH4 and O3), was used to determine greenhouse gases that released in the air due to burning fossil fuels including gasoline combustion from cars. As a result, it appears that Darwaza gas station gasoline (regular and premium), has the highest octane rating as compared to the others which makes it a better one to choose among them by utilizing additives like ethanol and (MTBE) to increase octane value. So, this means less environmental impact than PB but it's still harmful and banned in most of the countries in the world. on human due to air pollution by toxic gases that will release as a consequence of burning gasoline.

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List of Abbreviations

PB	LEAD	SUV	SPORT UTILITY VEHICLE
RVP	REID VAPOR PRESSURE	CO_2	CARBON DIOXIDE
RON	RESEARCH OCTANE NUMBER	NOX	NITROGEN OXIDE
MON	MOTOR OCTANE NUMBER	VOC	COLATILE ORGANIC COMPOUND
RV	RECREATIONAL VEHICLE	EPA	ENVORNMENTAL PROTECTION AGENCY
EV	ELECTRICITY VEHICLE	OPEC	THE ORGANIZATION OF PETROLEUM EXPORTING COUNTRIES
PM	PARTICULAR MATTER	PBL	PLANETARY BOUNDARY LAYER
LPG	LIQUEFIED PETROLEUM GAS	AKI	ANTI-KNOCK INDEX
MTBE	METHYL TERT-BUTYL ETHER	IBP	INITIAL BOILING POINT
FBP	FINAL BOILING POINT	PPM	PARTS PER MILLION
NO_x	NITROGEN OXIDES	NO_2	NITROGEN DIOXIDE
CO	CARBON MONOXIDE	NO	NITROGEN OXIDE
R	Regular	P	Premium

Chapter One

Introduction

1 Introduction

Gasoline is a type of fuel with a complex mixture of hydrocarbon compounds derived from petroleum.it is primarily composed of aliphatic hydrocarbons, such as pentane, hexane, heptane, and octane. These hydrocarbons undergo refining process to remove impurities and optimize their combustion properties. Gasoline is a volatile liquid that serves as a fuel for internal combustion engines in vehicles. There are different types of gasoline, such as regular, mid-grade, and premium. They have different octane ratings, which measure the fuel's ability to resist knocking in the engine. Regular gasoline has an octane rating of 87, mid-grade is around 89-90, and premium can be 91 or higher. In this study we distinguish between regular gasoline and premium gasoline in different petroleum stations in Sulaymaniyah city to analyses them according to their octane number (MON,RON), and extracting other components that made up the gasoline like (AKI, Benzene, Aromatics, Toluene, MTBE,Oxygen, Ethanol, Oxygenate, Olefins, IBP, FBP, T10, T50, T90) to understand the difference amount of these component in each station and deciding which one is best choice to use by having minimal impact on vehicles and has less environmental impact in Sulaymaniyah area. we used different device for this purpose like (Eralitex Vapor) for determining vapor presser in gasoline, and (Eralytics) or (fuel tester) that determine all of the other components in gasoline by using the latest modern of this device which is called (Zeltex).

1.1 History of Gasoline

various forms of gasoline have been used for several centuries, in September 5, 1885 the first gasoline pump was manufactured in Fort Wayne, Indiana. This was initially for tractors but was the first step towards the vast array of stations nationwide we have today, so gasoline was not invented, it was discovered accidentally as a waste by-product of the

petroleum refining to obtain kerosene for illumination, It wasn't until people started refining crude oil on a large scale that gasoline started being produced in significant quantities. This began with the drilling in 1859 of the Drake Well in Pennsylvania by Edwin Drake. The Drake Well is notable for being one of the first wells to be drilled solely for producing oil, and it did a lot to promote the use of oil-based products as fuel. Since people were originally refining crude oil solely to make kerosene, the gasoline that was produced during the refining process was deemed to have no use. As a result, early gasoline was usually converted into other types of fuel or just discarded outright. Kerosene being the principal product. Gasoline is produced by distillation, the separating of the volatile, more valuable fractions of crude petroleum. However, what was invented were the numerous processes and agents needed to improve the quality of gasoline making it a better commodity, gasoline started to be used as a solvent for general purposes. The first important leap to gasoline for automotive use was made by German engineer Karl Friedrich Michael Vaillant Benz , a graduate of the Karlsruhe Institute of Technology, who built the first gasoline fueled mono cylindric motor prototype in 1879, which finally was patented by Daimler in 1886. On June 11, 1895, the first U.S. patent for a gasoline-powered automobile was issued to Charles Duryea of Springfield, Massachusetts.

1.2 How Gasoline made

Gasoline is made by breaking down crude oil into petroleum products through fractional distillation, after which the usable gasoline is transported to gas stations via pipelines. the product is called Naphtha from the top of the Distillation Tower, The initial process, thermal reforming, was developed in the late 1920s. Thermal reforming employed temperatures of 510–565 °C (950–1,050 °F) at moderate pressures—about 40 bars (4 MPa), or 600 psi—to obtain gasolines (petrol) with octane numbers of 70 to 80 from heavy naphtha with octane numbers of less than 40. There are four major types of reactions that occur during reforming processes:

(1) dehydrogenation of naphthene to aromatics,

(2) dehydrocyclization of paraffins to aromatics,

(3) isomerization,

(4) hydrocracking

As of today, gasoline is among the most commonly used petroleum products, comprising approximately half of the products in use.

Gasoline is something individuals invest in on a regular basis. Whether you're part of a fleet or you simply use your car to get to and from work, you need gasoline to make your engine run. Heat, oxygen and humidity all have an impact on the condition of stored fuel.

1.3 Obtaining Crude Oil

Crude oil (also known as petroleum) is a black liquid composed of hydrocarbons and organic compounds. It's pumped out of the ground and can be transported or imported from another country in order to be turned into gasoline. The most common method for extracting crude oil from the ground is drilling into the porous rock and pumping it out using piping and valving. Once it's pumped out, it's transported to a petroleum refinery.

1.4 Distillation and Refining

Every day, 647 million gallons of oil are refined. This takes place at the large, multi-faceted processing plants called oil refinery, Gasoline also known as petrol which is an energy-dense secondary fuel that can be thought It is used to power many heat engines, most importantly it acts as a fuel for a large proportion of cars. Because crude oil requires a very high temperature to burn, it doesn't make a very good fuel and needs to be separated into smaller chains of refined fuel (including gasoline). Once the oil is at the refinery, it's pumped into a high-temperature furnace that evaporates everything but the largest molecules. The vapors travel into column and cool at different temperatures. Lighter products like butane, kerosene, and gasoline travel to the top of the column while the molecules at the bottom are removed and used for plastics and lubricants manufacturing. About 44% of petroleum will become gasoline, 22% will become diesel fuel, 9% will become jet fuel, and the rest will be used in a variety of other ways. Many times, petroleum is refined to maximize the output of its products and improve the quality of fuel. This requires a catalyst such as aluminum, platinum, acids, or processed clay to be added to the petroleum and then heated so that chemical changes occur and a purer gasoline product is created.

1.5 Additives and Octane Ratings

After refining gasoline, many companies add things like anti-knock compounds to prevent engine knocking or antioxidants to prevent the formation of resin in the engine. Once the gasoline is ready to use, it's evaluated for its mixture of heptane and isooctane and given a rating based on the ratio of the two liquids. The chemical composition of gasoline varies depending on its grade or octane rating, but generally speaking it is a mixture of combustible hydrocarbons. This octane rating describes the quality of the fuel, and the value is based on the ratios of two compounds in the gasoline - specifically iso-octane, a compound with the same chemical formula as octane but with a slightly different structure and properties, and normal heptane. The higher the amount of octane in the fuel, the larger the octane number and the higher the quality fuel. This higher quality of the fuel ensures that the ignition of the fuel happens on time as a result of a spark from the spark plug and not early as a result of compression from the piston. It is essential to running most internal combustion engine cars. Because of this, gasoline is one of the most widely used petroleum products. For example, 87 octane gasoline means there's 87% isooctane and 17% heptane in the mixture; 91 octane gasoline means there's 91% isooctane and 8% heptane. Because the isooctane evaporates slowly and produces virtually no knocking, some drivers prefer to use higher octane gasoline in their vehicles. The price of gasoline varies drastically all over the globe and this impacts the cost of running a vehicle. Additionally, the global economy has become increasingly more intertwined with oil production and prices thus impacting the consumer basket, More recently, gasoline is being mixed with a biofuel known as ethanol. In Canada, 87-Octane gasoline is allowed to consist of up to 10% ethanol, as this is the highest percentage of ethanol that a regular car engine can still function on. Additionally, the specific composition of gasoline results in a high energy density. This high energy density is what makes gasoline such a valuable fuel, as a relatively small volume of fuel can provide a large amount of useful energy. A primary function of gasoline additives is the modification of the combustion process so that vehicular emissions meet or exceed (EPA) requirements. For example, in cities with high smog levels, a

reformulated gasoline blend that burns more cleanly is required, and, during winter, fuel is oxygenated to reduce carbon monoxide formation. To achieve this, EPA requirements differ from area to area and season to season. Additives also play an important role in controlling the ignition characteristics of gasoline to ensure engines operate at their peak and to prevent uncontrolled fuel ignition that could cause engine damage.

1.6 Study Area

The area in which we have conducted the study is Suleimani, a city is in the north of Iraq and the gas stations that we have chosen are in the north souths and east of the city

1.7 The Aims of the Project

- Analyze components in premium and regular-grade gasoline from Sulaymaniyah city stations, focusing on octane number, AKI, Benzene, Aromatics, Toluene, MTBE, Oxygen, Ethanol, and more.
- Identify stations with high-quality gasoline to reduce environmental pollution.
- Evaluate the influence of gasoline quality on engine efficiency and emissions, considering additives and alternative compounds like ethanol.

1.8 Report layout (Research organize)

In Chapter One we have given a brief introduction about the topic in which we are going to conduct this research on.

In Chapter Two provides extra research that has been done before in the area for this purpose, which both talk about how using fossil fuel in cars, industries and other human activities can affect the quality of air in the region.

In Chapter Three consist of the theoretical topic of our work including sample collection from different stations and the station in Tanjero area.

In chapter Four we talked about the material and the methodology of our procedure and the devises and instruments that we have used to test our samples, and the result of our tests from the instruments, it also contains the result of greenhouse emission data from cars in different locations in Sulaymaniyah city.

In Chapter five we have concluded our subject and why we have done this project, we discussed the future plans that should be done to avoid environmental air pollution in a near future.

Chapter two

2 Literature Review

[1] also worked on air pollution and environmental problem in Sulaymaniyah city by taking samples from different location to determine some greenhouse gases that comes from car emission according to their models, industries, and burning fossil fuels. Using remote sensing and ground measurements.

[2] studied about air pollution and air particles in Sulaymaniyah area, by understanding the generation, transportation, and accumulation of aerosol particulate matter (PM) pollutants in the atmosphere that needs an investigation into its geographical and temporal fluctuations. Due to the absence of air quality monitoring stations and insufficient background data about PM issues in developing countries, remote sensing data was used because it has been widely utilized as an alternative to studying PM concentrations. Measurements from satellites and ground measurements introduce a complete overview of particle pollution behavior. The main aim of the study was to evaluate PM matter concentration for 14 locations over Sulaymaniyah city- Kurdistan region/Iraq from July to November 2021. The results of ground measurements showed high variability in PM concentration values and ranged between; 7-253, 8-273 and 11-169 µg m-3 for PM1, PM2.5 and PM10, respectively, among the studied locations during the study periods. MODIS predicted value for PM1, PM2.5 and PM10 ranged between 7.62-59.35, 6.64-43.65, and 50.54-62.17 µg m-3 , respectively. The main conclusion that can be identified from this study is; that remote sensing data utilized to study PM matter concentration; a dense ground monitoring network is required to improve monitoring and environmental services management in Sulaymaniyah City

Chapter three
Theory of the topic

3 Theory of the topic

3.1 Gasoline Properties

Gasoline is a complex mixture, and its properties can vary depending on factors such as the refining process, additives, and regional specifications. Here are some key properties of gasoline:

- **Chemical composition:** gasoline consists of hydrocarbons, which are organic compounds made up of hydrogen and carbon atoms. The specific hydrocarbons present in gasoline can vary but commonly include alkanes (paraffins), cycloalkanes (Naphthene's), and alkenes (Olefins).
- **Octane rating**: The octane rating is a measure of a fuel's resistance to knocking or pinging in an internal combustion engine. Higher octane ratings indicate a greater resistance to premature ignition. Gasoline is often labeled with an octane rating, such as regular (87 octane), mid-grade (89-90 octane), or premium (91-94 octane).
- **Vapor pressure:** Vapor pressure is a measure of how easily a liquid evaporates. Gasoline needs to vaporize for efficient combustion in an engine. However, excessive vapor pressure can contribute to vapor lock and other issues. Gasoline specifications often include limits on vapor pressure.
- **Density:** The density of gasoline may be estimated based on the types of hydrocarbons present. The specific mass and density of the fuel decrease with increasing hydrogen content in the molecule. This means that paraffins, naphthene's, and aromatics are listed first in increasing density order. Gasoline has a typical density of = 700-800 kg/m3 Gasoline has a lower density than diesel or other heavier

fuels. The density affects the energy content of the fuel and, consequently, its efficiency in terms of volume.

- **Distillation range:** Gasoline is a mixture of hydrocarbons with different boiling points. The distillation range of gasoline represents the range of temperatures at which various components vaporize during the refining process. Gasoline typically boils in the range of approximately 30 to 200 degrees Celsius (86 to 392 degrees Fahrenheit).
- **Flash point:** The flash point is the lowest temperature at which gasoline vapors can ignite when exposed to an open flame or spark. Gasoline is classified as a flammable liquid due to its relatively low flash point**.**
- **Color and odor:** Generally, the paraffinic hydrocarbons possess the mildest odor and the aromatics the strongest odor. The odor level is related to the chemical properties and volatility of the constituents. Odors due to the presence of sulfur compounds or unsaturated constituents are excluded by the specification because they are monitored separately. Normally, naphtha is colorless (water white), but naphtha containing higher amounts of aromatic constituents may be pale yellow. Colors like green, red or yellow are
- also added to differentiate the different grades of gasoline, Gasoline is typically colorless. Any color in the fuel may indicate the presence of impurities or additives.
- **Sulfur content**: The sulfur content in gasoline has been a significant focus due to environmental concerns. Low sulfur content is desirable to reduce emissions and prevent damage to catalytic converters.
- **Composition**: Gasoline composition has changed in parallel with the Spark Ignition gasoline engine development. Lead tetraethyl, PB (C2H5)4, was used as an additive from 1950 to 1995 as an anti-knocking agent. Further, benzene and aromatics are

also controlled within limits as per Euro specifications. Gasoline contains C4~C12 hydrocarbons.

- **Reid Vapor Pressure:** The Reid vapor pressure test is used to determine the front-end volatility of products in the gasoline through heavy reforming naphtha boiling point range. It correlates with the normal butane content of the sample and the RVP of gasoline blends is adjusted by adding or removing normal butane.
- **Volatility**: The evaporation of gasoline fuel in the intake channel and cylinder is affected by the fuel's volatility. When the fuel's volatility is low, a suitable air-fuel combination may be formed, but when the fuel's volatility is high, vapor bubbles can develop in the suction channel and block the flow of fuel as the local temperature increases.
- **flammability:** The flammable range of gasoline is the ratio of the flammable liquid to air that would create a volatile mixture. The flammability range of gasoline is between 1.4% and 7.6%. If the ratio of gasoline to air is less than 1.4%, then the mixture is too thin to burn.

3.2 Gasoline Types

Gasoline comes in various types and grades, often differentiated by their octane ratings and potential ethanol content. Internal combustion engines compress a mixture of air and fuel and ignite them with a spark. Under certain conditions, the fuel-air mixture can ignite early, creating a knocking sound in the engine. Newer cars have sensors that prevent pre ignition, so you'll rarely hear knocks and pings unless you drive an older vehicle.

A gasoline's octane rating averages two testing methods — the research octane number (RON) under idle conditions and the motor octane number (MON) under higher engine speed. You might have noticed this formula in smaller letters on the yellow buttons displayed on gas pumps — (R+M)/2.The larger numbers on gas pump stickers indicate the octane rating of the different gasoline grades. Typically, gas stations sell gasoline with three octane levels.

- **Regular:** Regular unleaded gasoline, typically labeled as 87 octane, is the most common type of gasoline. It is suitable for most vehicles and is the least expensive option. It meets the needs of the majority of cars on the road, Regular is a term given to the lowest octane fuel offered at gas stations. The "regular" designation is related to the cost of the fuel, which is directly related to the octane level. Most gas stations offer Regular, Mid-grade, and Premium fuels. Lower octane fuel such as Regular 87 is a less stable and burns unpredictably in high compression high-performance engines. 87 Octane does not provide the same power and performance as higher octane fuels. The instability of low octane fuels can result in uncontrolled or spontaneous ignition of the fuel in the engine cylinders resulting in significant damage to high-performance engines over time. Many cars using computer-controlled ignition that can adjust timing and sparking to limit spontaneous combustion, but it results in higher emissions and lower engine efficiency. Regular

gasoline is usually cheaper than the other grades or octanes of fuel sold at gas station pumps.

- **Mid Grade**: Mid-grade gasoline, often labeled as 89 or 90 octane, offers a higher octane rating than regular unleaded. Some vehicles, especially those with higher compression ratios, may benefit from using mid-grade fuel to prevent engine knocking, Mid-grade gasoline is typically 89 octane but can be 90 octane in certain regions. Certain cars, such as SUVs and modern pickup trucks, benefit from the higher octane fuel. The 89 octane benefits improved fuel efficiency, less wear and tear on the engine, and more power from the fuel. Most car manufacturers that recommend a mid-range fuel for their vehicles state that regular 87 grade can be used in these vehicles, but you cannot expect the same performance with this lower octane fuel. In most cases, vehicles that use 87 octane fuel can use 89 octane fuel without any damage, but there will not be the same performance increase as an engine designed for 89 octane. Using a higher octane gasoline in a car designed for a lower octane will not improve performance; you will simply be paying more for the gasoline.

- **Premium**: Premium unleaded gasoline has a higher octane rating, typically in the range of 91 to 94. It is designed for high-performance or luxury vehicles that require higher octane levels to prevent knocking or pinging, Even though some companies market their different grades of gas with names such as “Super Premium,” “Ultra,” or “Ultimate,” they all refer to the gasoline octane rating, Premium gasoline is typically between 91 and 93 octane, with the average being 92 octane. Premium gasoline is the most expensive fuel sold at gas station pumps, with mid-grade and regular-grade gasoline being cheaper.Premium octane fuel is more stable and limits the potential of pre-ignition or spontaneous ignition in the engine cylinder. The higher octane also gives a cleaner, more powerful burn than lower octane fuel.

3.3 Advantages and disadvantages of gasoline:

Gasoline, like any energy source, has both advantages and disadvantages. Here's an overview of the pros and cons of gasoline:

3.3.1 Advantages

Gasoline is primarily used as a fuel for internal combustion engines, powering various modes of transportation and certain equipment. Here are some of the primary uses of gasoline:

- **More common**: Gasoline-powered cars are more commonly used worldwide. Therefore, it is easier to access gasoline stations, which offers you more options while traveling or on long trips.
- **More affordable:** Gasoline-powered cars are generally more affordable than diesel-powered cars. The difference between rental fees and fuel costs can make a significant difference, especially for long-term car rentals.
- **Better performance:** Gasoline-powered cars offer higher performance due to their lighter and higher-revving engines. This advantage allows you to have better acceleration and torque, providing a more enjoyable driving experience.
- **Quieter:** Gasoline-powered cars provide a quieter driving experience compared to diesel-powered cars. This allows you to hear less noise while driving and provides a more comfortable driving experience.
- **Less maintenance**: Gasoline-powered cars require less maintenance compared to diesel-powered cars. This means lower maintenance costs and a lower risk of breakdowns. In addition, gasoline-powered cars usually have less complex systems, making them easier and cheaper to repair.
- **Fast Acceleration:** Gasoline accelerates much better than other fuels.
- **Quiet Combustion and Operation:** While the exhaust does contribute to

hydrocarbons in the atmosphere, it is highly favored for its quiet combustion and operation, particularly desired in passenger vehicles.

- **Versatility:** Gasoline can power a wide range of vehicles and equipment, from cars and motorcycles to lawnmowers and generators. This versatility contributes to its widespread use.
- **High Energy Density:** Gasoline has a high energy density, meaning it contains a significant amount of energy per unit of volume. This results in efficient energy storage and transportation.
- **Quick Starting:** Because it is highly combustible, vehicles powered by gasoline usually start up quicker than other vehicles.

3.3.2 Disadvantages

- **Emission is still among dreaded disadvantages of gasoline cars:** Even if Petrol vehicles are on average less polluting than diesel vehicles, they do emit carbon monoxide and other particulates into the environment upon combustion. Although the emission of fine particles and CO2 is less, there are around 1.3 billion cars out on the road worldwide, that many vehicular emissions seem too much to deteriorate the air quality.
- **Low fuel efficiency:** Gasoline cars have low fuel efficiency means that, even if you can avail fuel for your car easily, you won't be able to travel farther without refueling your car. This counts as one of the biggest disadvantages of gasoline cars as compared to hybrid and diesel counterparts.
- **Environmental Impact:** Burning gasoline produces carbon dioxide (CO2) and other pollutants, contributing to air pollution and climate change. Emissions from gasoline combustion are a significant environmental concern.
- **Air Pollution:** Gasoline combustion releases pollutants such as nitrogen oxides (NOx), particulate matter, and volatile organic compounds (VOCs), contributing to

smog and respiratory problems.

- **Gas-Powered cars have Lower Resale value:** Fuel efficiency and engine capacity of a gasoline car detriments over time. These factors directly affect the resale value of a vehicle as the same accounts for a limited lifespan.
- **Health Hazards:** Exposure to Gasoline can occur from filling your tank, from a gas leak in your engine, or from general spills and accidents. Every year, more than 5000 gas station fires are reported in the US alone, Humans can be affected by ingestion, inhalation, or even fire incidents at gas refilling stations. Gasoline cans reportedly ignited or exploded in about 5% of thermal burn injuries and fatalities.
- **Fossil Fuels Dependency:** It's a fact that Fossil fuels are non-renewable sources of energy, and gasoline is processed from these sources underneath the earth's surface. We may run out of gasoline in the nearing years, which is among the greatly feared disadvantages of gasoline cars and other vehicles. The value of gas-powered cars will plummet strikingly.
- **Noise and Vibrations**: Gasoline engines can produce noise and vibrations, which may be perceived as drawbacks in certain applications or for individuals seeking quieter and smoother alternatives.
- **Dependency on Oil:** The production and use of gasoline are linked to global oil markets, making economies vulnerable to fluctuations in oil prices and supply disruptions.
- **Finite Resource:** Gasoline is derived from crude oil, a finite and non-renewable resource. Concerns about depletion and geopolitical issues related to oil extraction and distribution are ongoing challenges.

3.4 Gasoline Additives

Gasoline additives are chemical compounds added to gasoline to enhance its performance, improve fuel efficiency, and reduce emissions. These additives serve various purposes, including preventing engine deposits, improving combustion efficiency, and meeting environmental standards.

3.4.1 Fuel additives:

Fuel additives work with the regular gas that you put in your tank. They function by, as the name suggests, adding substances, such as system cleaners, lubricants and performance enhancers, to the fuel to achieve a specific goal. No matter what your pump gasoline says or how it's advertised, it does not have anywhere near the additive packages a professionally formulated fuel treatment does.

People typically use additives in small quantities, and they're cheaper than changing the fuel itself. Modifications to an entire engine are obviously much more expensive, and additives can help drivers avoid them by addressing a specific problem or by helping with preventative maintenance. Using a fuel additive can solve problems on an engine without breaking the bank or the parts. Additives work on issues related to the movement of gasoline or the system it flows through. Cleaning agents, for instance, might be mixed with fuel to remove heavy deposits, reducing friction and improving flow. Individual components like injectors and valves need to be free of any buildup to work at their best. Cleaning out the system can put a stop to sluggish responses and hard starts, among other issues. Lubricants work similarly, helping parts to move better and more efficiently. Other tasks a fuel additive can take on include preventing gas from freezing, adding potentially performance-boosting octane and keeping fuel from going bad over time.

There are a variety of different types of fuel additives, including those meant for off-road and marine usage that may not be legal for use on the streets. Others might be intended for specific environments, such as in cold weather or high-moisture areas. Most fuel additives, however, are developed for an everyday car that might need a little something extra to get

3.4.2 Different types of fuel additives:

The benefits of fuel additives vary widely. They could be performance enhancers or fuel stabilizers. Fuel performance enhancers improve the functioning of your fuel. They may do so by cleaning a system and preventing buildup for the future. Fuel stabilizers keep the fuel intact, preventing adverse long-term effects, such as solidification and corrosion. Below are just some of the types of additives you can use in your vehicle.

1. **Fuel Stabilizers:** a fuel stabilizer will help your fuel hold up over time. Most stabilizers are intended for vehicles and equipment that sees intermittent use or goes long periods without seeing use. Seasonal vehicles such as boats and RVs, lawn care equipment and generators are a few examples. The additive improves efficiency by preventing the separation of gas and water. When fuel separates after being left sitting for a while, it can cause hard starts and engine corrosion. A stabilizer blocks this process and helps the components to stay thoroughly mixed. If your fuel will be sitting for over a month or so, it is probably a good idea to add some stabilizer, especially if conditions are cold.
2. **Octane Booster:** Before the ’70s, tetraethyl lead, high in octane, was used as an additive in gasoline. Of course, that practice ended when the health hazards of lead came to light. Some people still look for high-octane fuel to put in their cars, whether it makes sense or not. High-octane gasoline is often associated with high power, as it can increase the compression ratio. When the compression ratio is higher, there is more air in the engine’s cylinders. This creates a more powerful explosion when it ignites. The octane levels reflect how much the air-fuel mixture can compress before it ignites.
3. **Ethanol Treatment:** We hear about the benefits of ethanol gas all the time. Over 98% of gasoline in the US contains some ethanol, but that doesn’t mean it can’t have adverse effects on your car. Distributors use it for a variety of reasons, most notably

its widespread availability and capacity to reduce emissions of pollutants such as carbon monoxide. Ethanol fuel uses renewable sources, including corn, making it an attractive eco-friendly option. One problem with ethanol fuel is its tendency to absorb water from the air, making it more prone to separation. Also, some older engines aren't built for ethanol fuel and can have problems with deposit buildup.

We took samples from seven different petroleum stations in Sulaymaniyah city in different locations, which they are (Halgurd Petrol Station ,Sara Petrol Station ,Darwaza Petrol Station ,Qiwan Petrol Station , High Mountain Petrol Station ,Speed Fuel Petrol Station, Brayati petrol station). After that we analyzed them in gasoline test station in Tanjero area. Then data of air particles and gas emission like CO_2, NO_2, NO, O_3, CO were obtained from studies before because we couldn't afford or borrow the devise from anywhere in Sulaymaniyah city so we used data from [1]to understand what the quality of petroleum and car models can change the amount of gases in the atmosphere.

4 Material and Methodology

We took thirteen samples of premium and regular gasoline in multiple gas stations in Sulaymaniyah city like:

1- Halgurd Petrol Station

2- Sara Petrol Station

3- Darwaza Petrol Station

4- Qiwan Petrol Station

5-High Mountain Petrol Station

6-Speed Fuel Petrol Station

7-Brayati petrol station

In order to determine the quality of gasoline because as we all know the good and bad gasoline affects air pollution, human health, animals and the environment as well. We took the gasoline samples to the nongovernmental laboratory in Tanjero area, we used three types of equipment to test our samples, the first device is called the (Eralitex)

Vapor), which only tells us the vapor pressure of the gasoline, it is the latest device that costs about $ 60,000. It is easy to operate. The process started by putting a few milliliters of gasoline samples for each station through a pipe into the device.

The ERAVAP is the ultimate solution for the automated vapor pressure measurement of automotive and aviation gasoline, hydrocarbon solvents, crude oil and LPG as well as for the vapor/liquid ratio measurement of gasoline, the vapor pressure in winter must be in (50-77) , and in summer (44,59).

Figure 4.1(Eravap) vapor pressure tester.

The second device is (Eralytics) or (fuel tester) a completely advanced device.

Figure 4.2 (Eralytics) multi-fuel analyzer.

ERASPEC is eralytics' industry-proven multi-fuel analyzer with several thousand

installations worldwide. It offers measurements of fuel component concentrations, complex properties like RON or octane number, and density for gasoline, diesel fuel, jet fuel and even specialized fuels like ethanol fuels. A single push of a button delivers all results within 1 minute, and shows all the components that determine the quality of our premium and regular gasoline samples such as Ron, Mon, AKI, Benzene, Aromatics

Toluene, MTBE, and Oxygen, Ethanol, Oxygenate, Olefins, IBP, FBP, T10,T50, T90.

RON: The research octane number indicates the combustibility of engine fuel at low speeds and temperatures. It's designed to reflect the behavior of fuel under idling conditions and during acceleration. The higher the RON rating, the more compression it can withstand in a spark-ignition engine before igniting.

MON: The motor octane number denotes how a fuel behaves at full-throttle range. The rating is calculated at high speeds and temperatures designed to simulate fuel combustion on motorways and highways. When the MON rating is too low, the air/fuel mixture doesn't combust properly in the engine and creates a pinging sound known as "knocking". Optimizing fuel and engine pairings is one of the key uses of MON.

Aromatic: it must be in between 25-35

AKI: Anti-Knock Index.

AKI or (RON + MON)/2

Oxygenates: are fuel additives that contain oxygen, usually in the form of alcohol or ether. Oxygenates can enhance fuel combustion and thereby reduce exhaust emissions. Some oxygenates also boost gasoline octane should be in between 0-3.

- **Benzene:** 0-15
- **Olefin:** 0-18.

Toluene: is a gasoline additive that can be used to improve octane ratings for fuel used in race cars and cleaning machines should be in between 0-10

The higher the octane number or rating, the greater the fuel's resistance to knocking or pinging during combustion. **Methyl tert-butyl ether (MTBE):** is a flammable liquid that has been used as an additive for unleaded gasoline since the 1980s. MTBE increases octane and oxygen levels in gasoline and reduces pollution emissions. Because of concerns for groundwater contamination and water quality, MTBE is now banned or limited in several states. MTBE is also used in small amounts as a laboratory solvent and for some medical applications must be between 0-10.

Ethanol: is a renewable fuel made from various plant materials collectively known as "biomass.

The boilling point of the lightest component in a petroleum mixture is called initial boilling point (IBP) and the boiling point of the heaviest compound is called the final boiling point (FBP). In some references the (FBP) is also called the end point.

Temperatures for evaporation of 10 % (T10), 50% (T50), 90% (T90) and final boiling point (FBP) of fuel are important features of the distillation curve and their limits or equivalent are specified in the gasoline standard specifications in between 0-188 the higher the better.

The third device (Zeltex) or (octane tester) is a small, inexpensive device that can be easily transported but it only tells us the octane number of the gasoline but it is not accurate as much as the past

The forth devise is Portable gas analyzer-series 500 model with three sensor heads to measure CO2,CH4

and O3

Figure 4.3 (Zeltex) octane test

Figure 4.4 Portable gas analyzer-series 500

4.1 Results

After testing our thirteen premium and regular gasoline samples here are the outcome of our examinations.

Table 4.1Sample ID: R - Sara

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7363 G/CCM					
RON		-	-	93.3	-	MLR
MON		-	-	85.7	-	MLR
AKI		-	-	89.5	-	-
BENZENE		0.94	1.12	-	-	MLR
AROMATICS		29.2	34.6	-	-	-
TOLUENE		9.83	11.6	-	-	-
MTBE		5.10	5.14	-	-	-
OXYGEN		-	1.23	-	-	-
ETHANOL		0.80	0.86	-	-	-
OXGYNATES		5.90	6.00	-	-	-
OLEFINS		1.4	1.4	-	-	-
IBP		-	-	32.9	C	-
FBP		-	-	184.0	C	-
T10		-	-	47.2	C	-
T50		-	-	88.5	C	-
T90		-	-	149.5	C	-
Vapour pressure		-	-	58.4	KPa	-

Table 4.2 Sample ID : P- Sara oil

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	07476 G/CCM					
RON		-	-	95.4	-	MLR
MON		-	-	86.9	-	MLR
AKI		-	-	90.7	-	-
BENZENE		0.89	1.05	-	-	MLR
AROMATICS		30.0	35.0	-	-	-
TOLUENE		8.67	10.1	-	-	-
MTBE		6.39	6.34	-	-	-
OXYGEN		-	1.45	-	-	-
ETHANOL		0.81	0.85	-	-	-
OXGYNATES		7.19	7.20	-	-	-
OLEFINS		4.4	4.3	-	-	-
IBP		-	-	32.5	C	-
FBP		-	-	190.6	C	-
T10		-	-	49.2	C	-
T50		-	-	95	C	-
T90		-	-	157.9	C	-
Vapour pressure		-	-	55	KPa	-

Table 4.3 Sample ID : P - Darwaza

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7418 G/CCM					
RON		-	-	96.9	-	MLR
MON		-	-	87.6	-	MLR
AKI		-	-	92.3	-	-
BENZENE		0.52	0.62	-	-	-
AROMATICS		29.5	34.8	-	-	MLR
TOLUENE		14.5	17.0	-	-	-
MTBE		7.03	7.03	-	-	-
OXYGEN		-	1.52	-	-	-
ETHANOL		0.65	0.70	-	-	-
OXGYNATES		7.68	7.73	-	-	-
OLEFINS		-	-	-	-	-
IBP		-	-	34.0	C	-
FBP		-	-	184.0	C	-
T10		-	-	47.6	C	-
T50		-	-	96.7	C	-
T90		-	-	146.6	C	-
Vapour pressure		-	-	59.3	KPa	-

Table 4.4 Sample ID : R Darwaza oil

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7280 G/CCM					
RON		-	-	94.2	-	MLR
MON		-	-	87.2	-	MLR
AKI		-	-	90.7	-	-
BENZENE		0.51	0.62	-	-	MLR
AROMATICS		25.9	31.0	-	-	-
TOLUENE		9.91	11.8	-	-	-
MTBE		4.66	4.75	-	-	-
OXYGEN		-	0.86	-	-	-
ETHANOL		-	-	-	-	-
OXGYNATES		4.6	4.75	-	-	-
OLEFINS		-	-	-	-	-
IBP		-	-	32.5	C	-
FBP		-	-	182.1	C	-
T10		-	-	43.0	C	-
T50		-	-	84.5	C	-
T90		-	-	147.5	C	-
Vapour pressure		-	-	60.4	KPa	-

Table 4.5 Sample ID : R Halgurd

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7368 G/CCM	-	-	-	-	-
RON		-	-	93.1		MLR
MON		-	-	86.0	-	MLR
AKI		-	-	89.5	-	-
BENZENE		0.81	0.97	-	-	MLR
AROMATICS		29.5	34.9	-	-	-
TOLUENE		9.05	10.7	-	-	-
MTBE		4.01	4.04	-	-	-
OXYGEN		-	1.04	-	-	-
ETHANOL		0.81	0.88	-	-	-
OXGYNATES		4.83	4.92	-	-	-
OLEFINS		-	-	-	-	-
IBP		-	-	32.8	C	-
FBP		-	-	183.5	C	-
T10		-	-	48.6	C	-
T50		-	-	92.0	C	-
T90		-	-	153.1	C	-
Vapour pressure		-	-	59.3	KPa	-

Table 4.6 Sample ID:P - Halgurd

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7478G/CCM					
RON		-	-	95.4	-	MLR
MON		-	-	85.7	-	MLR
AKI		-	-	90.6	-	-
BENZENE		0.91	1.08	-	-	-
AROMATICS		30.0	35.1	-	-	MLR
TOLUENE		9.02	10.5	-	-	-
MTBE		6.97	6.91	-	-	-
OXYGEN		-	1.51	-	-	-
ETHANOL		0.70	0.74	-	-	-
OXGYNATES		7.66	7.65	-	-	-
OLEFINS		4.8	4.7	-	-	-
IBP		-	-	32.4	C	-
FBP		-	-	190.5	C	-
T10		-	-	50.1	C	-
T50		-	-	96.3	C	-
T90		-	-	158.0	C	-
Vapour pressure		-	-	54	KPa	-

Table 4.7 Sample ID : R – High Mountain

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7400 G/CCM					
RON		-	-	93.1	-	MLR
MON		-	-	86.0	-	MLR
AKI		-	-	89.5	-	-
BENZENE		0.74	0.88	-	-	MLR
AROMATICS		30.8	36.3	-	-	-
TOLUENE		10.5	12.4	-	-	-
MTBE		-	-	-	-	-
OXYGEN		-	0.45	-	-	-
ETHANOL		1.21	1.29	-	-	-
OXGYNATES		1.21	1.29	-	-	-
OLEFINS		0.9	0.9	-	-	-
IBP		-	-	32.9	C	-
FBP		-	-	181.5	C	-
T10		-	-	48.3	C	-
T50		-	-	91.9	C	-
T90		-	-	151.5	C	-
Vapour pressure		-	-	54.1	KPa	-

Table 4.8 Sample ID : P - High Mountain

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7498 G/CCM					
RON		-	-	95.9	-	MLR
MON		-	-	86.9	-	MLR
AKI		-	-	91.4	-	-
BENZENE		0.70	0.82	-	-	-
AROMATICS		32.8	38.2	-	-	MLR
TOLUENE		10.8	12.5	-	-	-
MTBE		4.39	4.35	-	-	-
OXYGEN		-	1.20		-	-
ETHANOL		1.10	1.17	-	-	-
OXGYNATES		5.50	5.52	-	-	-
OLEFINS		0.6	0.6	-	-	-
IBP		-	-	34.6	C	-
FBP		-	-	184.0	C	-
T10		-	-	50.3	C	-
T50		-	-	92.5	C	-
T90		-	-	153.6	C	-
Vapour pressure		-	-	57.5	KPa	-

Table 4.9 Sample ID : R - Qaiwan Group

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7151 G/CCM					
RON		-	-	90.7	-	MLR
MON		-	-	85.1	-	MLR
AKI		-	-	87.9	-	-
BENZENE		0.40	0.49	-	-	MLR
AROMATICS		21.9	26.7	-	-	-
TOLUENE		9.00	11.0	-	-	-
MTBE		-	-	-	-	-
OXYGEN		-	-	-	-	-
ETHANOL		-	-	-	-	-
OXGYNATES		-	-	-	-	-
OLEFINS		-	-	-	-	-
IBP		-	-	32.2	C	-
FBP		-	-	179.1	C	-
T10		-	-	44.2	C	-
T50		-	-	85.8	C	-
T90		-	-	146.1	C	-
Vapour pressure		-	-	64.2	KPa	-

Table 4.10 Sample ID : P - Qaiwan Group

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7468 G/CCM					
RON		-	-	95.4	-	MLR
MON		-	-	86.7	-	MLR
AKI		-	-	91.1	-	-
BENZENE		0.85	1.00	-	-	MLR
AROMATICS		30.1	35.2	-	-	-
TOLUENE		9.51	11.1	-	-	-
MTBE		6.02	5.98	-	-	-
OXYGEN		-	1.39	-	-	-
ETHANOL		0.81	0.86	-	-	-
OXGYNATES		6.83	6.85	-	-	-
OLEFINS		1.2	1.2	-	-	-
IBP		-	-	33.8	C	-
FBP		-	-	186.3	C	-
T10		-	-	48.4	C	-
T50		-	-	91.5	C	-
T90		-	-	154.8	C	-
Vapour pressure		-	-	55.1	KPa	-

Table 4.11 Sample ID : R - Brayaty (750IQD)

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7143 G/CCM					
RON		-	-	95.4	-	MLR
MON		-	-	86.7	-	MLR
AKI		-	-	91.1	-	-
BENZENE		0.32	0.40	-	-	MLR
AROMATICS		22.9	28.0	-	-	-
TOLUENE		8.92	10.9	-	-	-
MTBE		-	-	-	-	-
OXYGEN		-	-	-	-	-
ETHANOL		-	-	-	-	-
OXGYNATES		0	0	-	-	-
OLEFINS		0	0	-	-	-
IBP		-	-	30.8	C	-
FBP		-	-	180.8	C	-
T10		-	-	43.2	C	-
T50		-	-	86.9	C	-
T90		-	-	146.5	C	-
Vapour pressure		-	-	65.0	KPa	-

Table 4.12 Sample ID : R - Speed Fuel

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7145 G/CCM					
RON		-	-	91.3	-	MLR
MON		-	-	85.8	-	MLR
AKI		-	-	88.5	-	-
BENZENE		0.66	0.81	-	-	MLR
AROMATICS		23.3	28.5	-	-	-
TOLUENE		7.51	9.15	-	-	-
MTBE		2.00	2.07	-	-	-
OXYGEN		-	0.72	-	-	-
ETHANOL		0.88	0.97	-	-	-
OXGYNATES		2.88	3.05	-	-	-
OLEFINS		-	-	-	-	-
IBP		-	-	31.8	C	-
FBP		-	-	180.4	C	-
T10		-	-	44.0	C	-
T50		-	-	83.4	C	-
T90		-	-	148.1	C	-
Vapour pressure		-	-	64.5	KPa	-

Table 4.13 Sample ID : P Speed Fuel

		VOL %	MASS%	VALUE	UNIT	TYPE
Density	0.7421 G/CCM					
RON		-	-	93.8	-	MLR
MON		-	-	86.6	-	MLR
AKI		-	-	90.2	-	-
BENZENE		0.81	0.96	-	-	MLR
AROMATICS		32.6	38.4	-	-	-
TOLUENE		10.3	12.1	-	-	-
MTBE		1.47	1.47	-	-	-
OXYGEN		-	0.64	-	-	-
ETHANOL		1.02	1.08	-	-	-
OXGYNATES		2.48	2.55	-	-	-
OLEFINS		0.6	0.6	-	-	-
IBP		-	-	32.5	C	-
FBP		-	-	182.2	C	-
T10		-	-	48.1	C	-
T50		-	-	93.0	C	-
T90		-	-	152.7	C	-
Vapour pressure		-	-	53.4	KPa	-

Table 4.14 Result of the premium gasoline from stations.

Stations	RON value
1. Darwaza oil	96.9
2. High mountain Oil	95.9
3. Halgord oil	95.4
4. Sara oil	95.4
5. Qaiwan group oil	95.4
6. Speed fuel oil	93.8

Table 4.15 Result of regular gasoline from stations

Stations	RON value
1. Darwaza oil	94.2
2. Sara oil	93.3
3. Halgord oil	93.1
4. High mountain oil	93.1
5. Speed fuel oil	91.3
6. Brayati oil	91.1
7. Qaiwan group oil	90.7

Results of measuring gas concentration that released from vehicle due to gasoline combustion from eight different locations in Sulaymaniyah city.

Table 4.16 Ranges percent of exhaust emitted CO2 with corresponded vehicle number[1]

NO	Location	Summrized values of 7 Reading	O_2%	Gases Concentrations (ppm)			Gases Concentrations (ppb)		
				CO_2	N_2O	CO	NO	NO_2	O_3
1-	Raparin / At sulaimani-Karkuk street	Average	21.0	402.3	28.0	2.8	1.4	2.1	35.9
		Normal	18.6	365.0	3.0	ND	ND	ND	5.0
		Maximum	23.1	494.0	58.0	11.0	20.0	10.0	69.0
2-	Sarchnar crossing	Average	20.8	477.0	26.4	5.6	27.9	10.0	40.6
		Normal	18.7	385.0	5.0	ND	ND	2.9	14.0
		Maximum	23.3	676.0	62.0	12.5	140	ND	76.0
3-	Wluba overpass	Average	20.6	394.9	70.3	24.1	2.2	23.6	6.4
		Normal	18.6	365.0	56.0	4.0	ND	ND	ND
		Maximum	22.7	487.0	86.0	67.0	8.3	190.0	20.0
4-	Tnajaro/near to tanjaro mosque	Average	20.5	430.2	24.8	2.1	56.4	2.1	34.6
		Normal	18.9	365.0	5.0	ND	ND	ND	3.0
		Maximum	22.8	700.0	62.0	8.7	400.0	10.0	63.0
5-	Foothill of goizha Mountain	Average	20.9	371.3	12.4	0.2	ND	0.7	28.9
		Normal	19.1	347.0	4.0	ND	ND	ND	5.0
		Maximum	22.5	444.0	39.0	0.8	ND	10.0	51.0
6-	Outside Peshraw Tunnel/About 75m far from the north side	Average	20.3	385.3	11.5	0.8	7.1	1.4	25.6
		Normal	18.1	359.0	1.0	ND	ND	ND	5.0
		Maximum	21.9	487.0	39.0	5.7	70.0	10.0	69.0
7-	Inside peshraw Tunnel/About 75m far from northside	Average	20.0	1159.1	22.1	5.3	54.3	17.1	124.7
		Normal	17.7	426.0	3.0	ND	ND	ND	8.0
		Maximum	22.1	2311.0	43.0	20.8	200.0	60.0	432.0
8-	Maleek Mahmood Circle/Opposite Binai Station	Average	20.5	503.9	22.6	4.0	59.3	7.1	54.4
		Normal	17.7	371.0	6.0	ND	ND	ND	14.0
		Maximum	22.1	1152.0	46.0	7.6	260.0	20.0	12.0

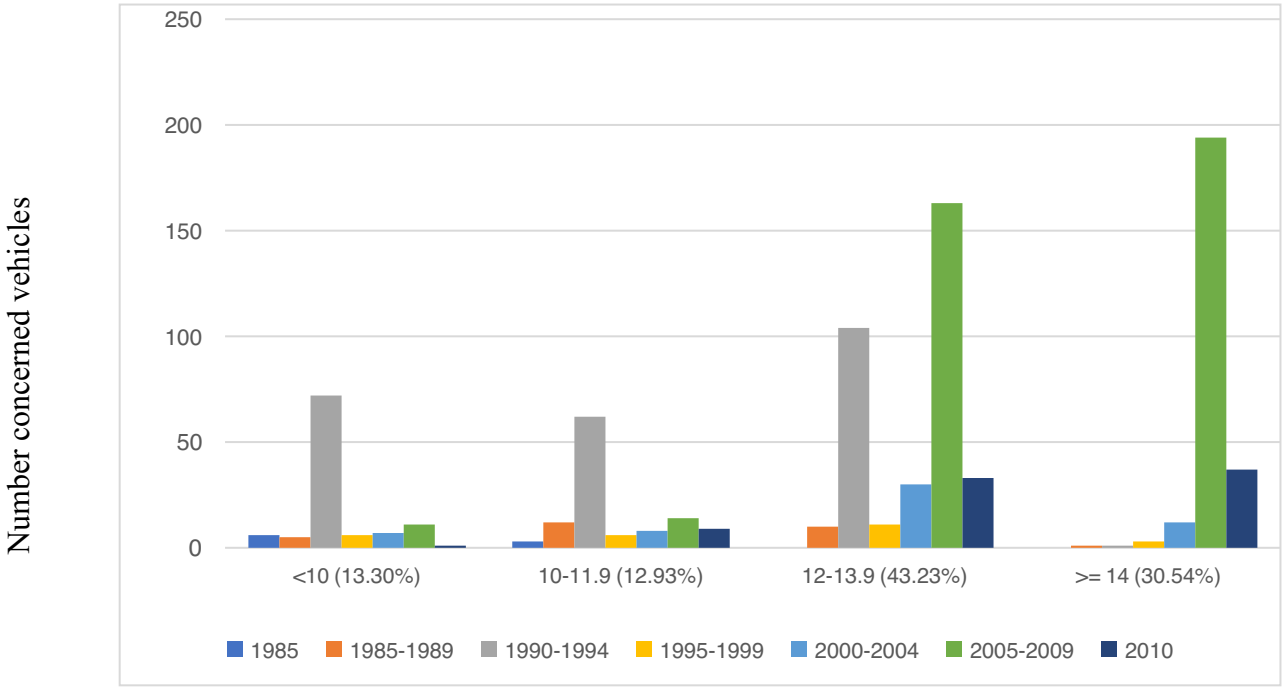

Table 4.17 Spatial distribution of CO2 concentrations (ppm) in July 2022 ground measurements for 13 locations [3].

	1985	1985-1989	1990-1994	1995-1999	2000-2004	2005-2009	2010
<10 (13.30%)	6	5	72	6	7	11	1
10-11.9 (12.93%)	3	12	62	6	8	14	9
12-13.9 (43.23%)	0	10	104	11	30	163	33
>= 14 (30.54%)	0	1	1	3	12	194	37

Chapter Five

Conclusion and Discussion

5 Conclusion

Through this study we examined premium and regular gasoline from different locations in Sulaymaniyah city and tested in Tanjero area by three different instruments to determine different components in them, as a result its shown that Darwaza oil station has the best quality than other stations that we tested due its high-octane rate and other components. increasing octane value is one of the procedures that has been done recently in the world by adding different additives like Lead (PB) which is banned now in the world due its harmful environmental impact, but unfortunately, it's still available in Iraq and a few other countries. In Kurdistan region and Sulaymaniyah specifically, gas stations use MTB to increase octane rating which a little better than (PB), Ethanol is a better alternative of (PB) and MTB which fortunately used by gas stations in Sulaymaniyah city along with MTB to raise the RON of gasoline which result to a better combustion for vehicles and less toxic greenhouse emission. but this cannot be reliable because emission of CO_2 and other greenhouses still occur due to burning fossil fuels specially from cars if serious strategies have not taking by government. As its shown from [1],[3] from these studies the amount of CO_2 and other greenhouse gases that released from burning gasoline increased dramatically in different places and it's the highest a mount in those place where there are traffic of cars depending on their models like in Peshraw Tunnel.

6 Discussion

6.1 The Future of Gasoline

Despite the increasing popularity of electric vehicles (EVs), gasoline is still critical to the global market and has no end yet in sight. Until sustainable options are both affordable and convenient, consumers will continue relying on gasoline-powered vehicles for transportation. However, as the EV market grows and develops, the demand for gasoline may slow. Unfortunately, the Organization of Petroleum Exporting Countries (OPEC) takes advantage of this. To ensure they receive a high price per barrel, they inflate prices by reducing production. In the end, the future of gasoline will hinge upon environmental, economic and technological factors and developments. For choosing the better quality grade between premium and regular grade gasoline, Premium-grade gasoline contains the same ethanol content and is no more powerful or quality than mid-grade fuel. It's simply more detonation-resistant. This may be required for certain vehicles like luxury cars, high-performance trucks, and sports cars. Those vehicles need gasoline to withstand higher temperatures without premature detonation so the engine doesn't sustain damage.

6.2 Environmental impact

Burning gasoline has a very bad impact on the environment and humans Because gasoline is made from several chemical compounds such as Aromatic, M T B E, PB, Sulphur, olefine, toluene, Ethanol, All of these, when burned by car engines, cause climate change and cancer. For example, to burn two moles of octane requires twenty-five moles of oxygen to keep the environment clean, PB is the main cause of cancer. This substance is allowed in only five countries, including Iraq. The amount of soot particles produced per liter of gasoline should not exceed five mlg. But in Sulaymaniyah, five grams per liter of cars without exhaust stones, So you have to stay away from areas where there are a lot of cars to be healthy, Also (MTBE) is a completely toxic ingredient, the standard rate should not

exceed 1%. Currently, this ingredient is banned in any way in the world.

CO_2 emissions from cars cause global warming and respiratory disease in humans which is the standard rate of 423 ppm,1percentage=10000ppm.

To reduce the impact of these components on the environment a lot of projects should be done like, the amount of greenery should be increased and reducing the use of bras of all types of fuels.

Ways to prevent car smoke in general and reduce the damage to the environment and health: First, Less driving: Less driving means less emissions. In order to use the car for less time, these steps are important to follow.

- Use bicycles and walk and avoid continuous use go to the market and daily work.
- Use public transportation if available.
- Sharing transportation with close people to markets and public places.
- Arrange to go out and gather several things and do them at the same time.
- Second, drive wisely. Minimum emissions and minimum expenditure:-
- Regularly maintain and repair your vehicle.
- Use the right oil for the vehicle.
- While driving, you can not suddenly increase or decrease the pressure on the gasoline and brake systems, thus reducing fuel consumption and increasing the safety and life of the engine.

Third, use the right quality fuel for the vehicle according to the type of engine.

Fourth: Heating the car: Avoid heating during operation for a long time, which causes emissions of pollutants and waste of fuel and expenses, try to use high quality cars that do not need to be heated, especially in winter.

Fifth: Electric vehicles: One of the mechanical and scientific inventions that are environmentally friendly as a step to protect the environment and reduce pollutant emissions into the atmosphere and adapt to the steps to reduce global warming ((Global Warming). There are about (18) million cars in the world, which is 2.2% of the total number of cars in the world. This type of vehicle is completely environmentally friendly because the energy source is electricity and does not use any fuel, so there are no emissions into،

7 Bibliography

[1] Majid, S, N. Valuation of ambient air pollution: a study of some urban areas in sulaimani city and its surrounding.107, 2011.

[2] Mohammed, Y, H.,Majid, N, S. Ambient Particulate Matter Concentrations for Difference Size from MODIS Satellite Images andGround Measurements in Sulaimani, IRAQ. Applied Ecology and Environmental Sciences. 10, 622-639. 2022.

[3] Abdalrahman, J, W. Assessment of some greenhouse gases in sulaimani governorate using remote ensing and ground measurements.21-196.2022.

[4] Gas Station Pump Five Octane Ratings" by Original uploader was Bobak at en.wikipedia - Transferred from en.wikipedia; transfer was stated to be made by User:Matt.T.. Licensed under CC BY-SA 2.5 via Wikimedia Commons - http://commons.wikimedia.org/wiki/File:Gas_Station_Pump_Five_Octane_Ratings.jpg#mediaviewer/File:Gas_Station_Pump_Five_Octane_Ratings.jpg

[5] Petro-Canada.2015. *Octane & Other Gasoline Basics* [Online]. Available: http://retail.petro-canada.ca/en/independent/2069.aspx

[6] Thomas, George. Overview of Storage Development DOE Hydrogen Program. Livermore, CA. Sandia National Laboratories. 2000.

[7] Tony Greenham., 2015. *The economics of oil dependence: a glass ceiling to recovery* [Online]. Available: http://www.neweconomics.org/publications/entry/the-economics-of-oil-dependence-a-glass-ceiling-to-recovery.

[8] U.S. Energy Information Administration (EIA).2024. Monthly Energy Review.221,222.

[9] Scotte, A. Pickard, S.oil and gas, poverty, the environment and human rights,2023.FAQ 3: oil and gas, poverty, the environment and human rights | ODI: Think change

[10] Gibbs, L., "How Gasoline Has Changed," SAE Technical Paper 932828, 1993, https://doi.org/10.4271/932828.

[11] Bellis, M. 2019.History of gasoline. History& Culture. History and Invention of Gasoline (thoughtco.com)

[12] Allison, E. Mandler, B. Petroleum and the environment.AGI. 2018.